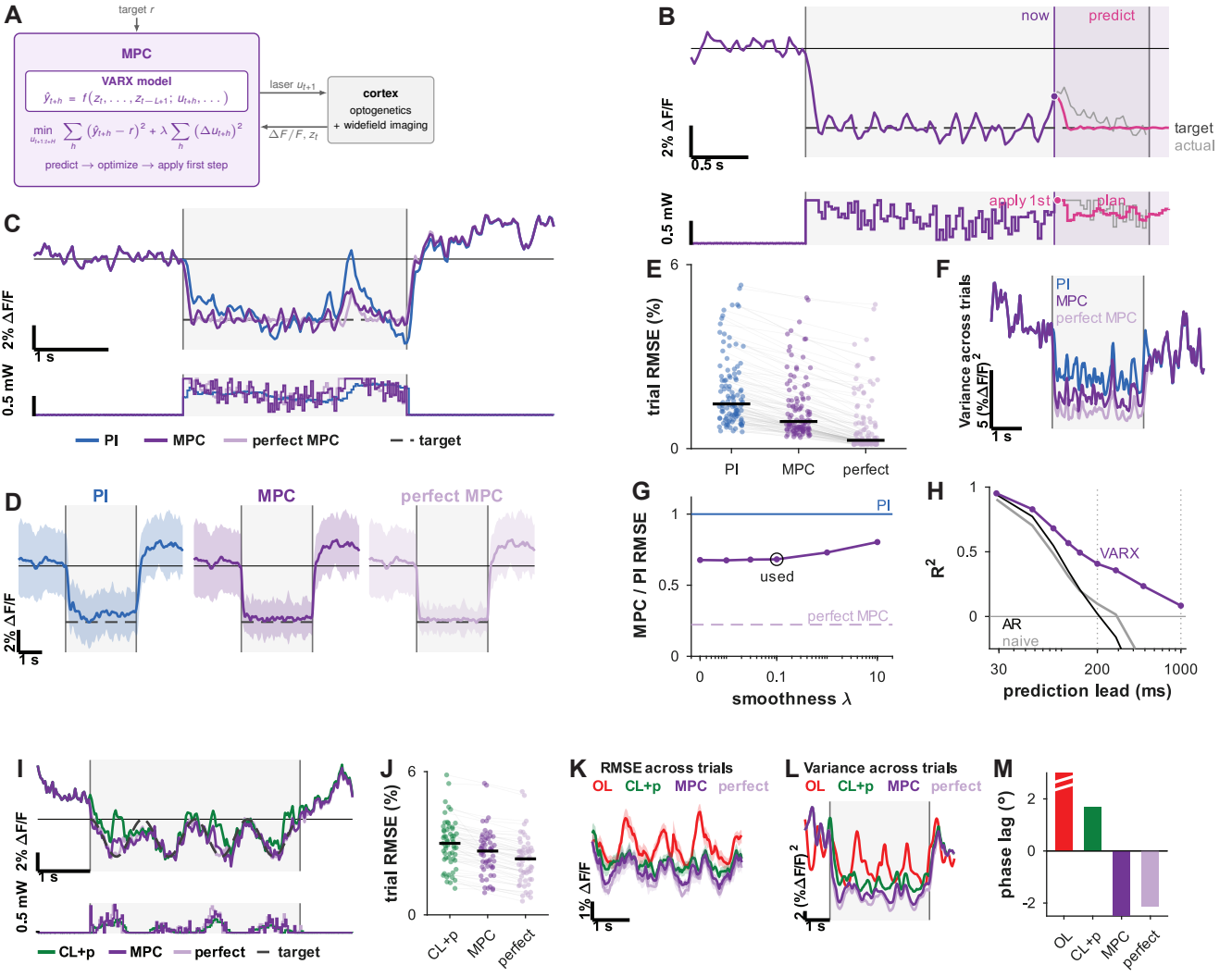

A
target r
MPC
VARX model
$\hat{y}_{t+h} = f(z_t, \ldots, z_{t-L+1}; u_{t+h}, \ldots)$
$\min_{u_{t+1:t+H}} \sum_h (\hat{y}_{t+h} - r)^2 + \lambda \sum_h (\Delta u_{t+h})^2$
predict → optimize → apply first step
laser u_{t+1}
$\Delta F/F, z_t$
cortex
optogenetics
+ widefield imaging
B
now
predict
2% ΔF/F
0.5 s
target
actual
0.5 mW
apply 1st
plan
C
2% ΔF/F
1 s
0.5 mW
PI
MPC
perfect MPC
target
D
PI
MPC
perfect MPC
2% ΔF/F
1 s
E
trial RMSE (%)
6
0
PI
MPC
perfect
F
PI
MPC
perfect MPC
Variance across trials
5 (%ΔF/F)^2
1 s
G
MPC / PI RMSE
1
0.5
0
PI
used
perfect MPC
0
0.1
10
smoothness λ
H
R^2
1
0.5
0
VARX
AR
naive
30
200
1000
prediction lead (ms)
I
2% ΔF/F
1 s
0.5 mW
CL+p
MPC
perfect
target
J
trial RMSE (%)
6
0
CL+p
MPC
perfect
K
RMSE across trials
OL
CL+p
MPC
perfect
1% ΔF/F
1 s
L
Variance across trials
OL
CL+p
MPC
perfect
2 (%ΔF/F)^2
1 s
M
phase lag (°)
2
0
-2
OL
CL+p
MPC
perfect